

Security Linter Report

Filename	case11_enterprise_app.py
Timestamp	2026-05-04 04:00:57 UTC
Lines of Code (LOC)	50
Verdict	VULNERABLE

ID	Severity	Sink	Source Variable	Source Type	Line
CVE-SL-002	WARNING	cursor.execute	params	EXTERNAL	71
CVE-SL-003	WARNING	cursor.execute	params2	EXTERNAL	72

CRITICAL	0
WARNING	2
INFO	0
TOTAL	2

CVE-SL-002 — cursor.execute**WARNING**

CWE	CWE-89
Sink	cursor.execute
Sink Line	71
Sink Column	0
Sink Expression	cursor.execute(query, params)
Argument	params
Source Variable	params
Source Type	EXTERNAL
Propagation Mechanism	data flow propagation

Taint Path Trace:

Step	Role	Line	Expression
1	SOURCE	41	params = (col, table, col, safe_value)
2	SINK	71	cursor.execute(query, params)

Remediation:

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

CVE-SL-003 — cursor.execute**WARNING**

CWE	CWE-89
Sink	cursor.execute
Sink Line	72
Sink Column	0
Sink Expression	cursor.execute(query2, params2)
Argument	params2
Source Variable	params2
Source Type	EXTERNAL
Propagation Mechanism	data flow propagation

Taint Path Trace:

Step	Role	Line	Expression
1	SOURCE	49	params2 = (safe_category,)
2	SINK	72	cursor.execute(query2, params2)

Remediation:

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

Sink	Line	Sanitizer	Protected Variable
cursor.execute	71	int	safe_new
cursor.execute	71	int	safe_cond

Function	Variable	Line
int()	safe_new	N/A
int()	safe_cond	N/A

CFG — Total Nodes	45
CFG — Condition/Loop Nodes	2
CFG — Assignment Nodes	23
CFG — Call Nodes	6
CFG — Merge Nodes	5
██	██████████████
DFG — Total Nodes	86
DFG — Total Edges	90
DFG — Source Nodes	7
DFG — Sink Nodes	3
DFG — Operator Nodes	5
DFG — Constant Nodes	14
DFG — Variable Nodes	37
Taint Propagation Hops	33
██	██████████████
Analysis Time (ms)	198.7

Vulnerabilities	0 CRITICAL 2 WARNING 0 INFO
Safe Sinks	2
Analysis Time	198.7 ms
Engine Version	1.0.0
Final Verdict	VULNERABLE